
(a) Header bar: simulation clock and time control, alarm banner
EmuBAS
DOE Reference Large Office (Chic…
15:05
Fri 14 Jul 2023
PAUSED
SPEED
1×
2×
10×
60×
360×
Max
Resume
Step
21 active alarms - AHU-4: Supply fan commanded on but status off (+20 more)
ALARMS 21
INJECTIONS 11
(b) Building section (zone tiles per floor)
Building section zone air temperature vs. heating / cooling set points
below heating SP
in band
above cooling SP
Floor 12
top floor
AHU-3
South · P1
22.8 °C
E+ 22.6
SP 21.0 – 23.0 °C
CO₂ 405 ppm
Flow 1.30 m³/s
OCCUPIED 14 p
East · P2
23.8 °C
E+ 23.6
SP 21.0 – 24.0 °C
CO₂ 496 ppm
Flow 0.57 m³/s
OCCUPIED 11 p
North · P3
24.0 °C
E+ 23.6
SP 21.0 – 24.0 °C
CO₂ 473 ppm
Flow 0.83 m³/s
OCCUPIED 14 p
West · P4
24.4 °C
E+ 24.2
SP 21.0 – 24.0 °C
CO₂ 465 ppm
Flow 0.97 m³/s
OCCUPIED 10 p
Core
24.0 °C
E+ 24.0
SP 21.0 – 24.0 °C
CO₂ 555 ppm
Flow 3.77 m³/s
OCCUPIED 110
Floors 2–11
mid floors, modelled once ×10
AHU-2
South … ×10
23.8 °C
E+ 23.6
SP 21.0 – 24.0 °C
CO₂ 753 ppm
Flow 7.62 m³/s
OCCUPIED 17 p
East · P2 ×10
22.2 °C
E+ 22.4
SP 21.0 – 24.0 °C
CO₂ 735 ppm
Flow 5.68 m³/s
OCCUPIED 10 p
North … ×10
24.0 °C
E+ 23.9
SP 21.0 – 24.0 °C
CO₂ 760 ppm
Flow 5.20 m³/s
OCCUPIED 17 p
West · … ×10
24.2 °C
E+ 24.3
SP 21.0 – 24.0 °C
CO₂ 711 ppm
Flow 7.57 m³/s
OCCUPIED 11 p
Core ×10
24.1 °C
E+ 20.3
Δ+3.84
SP 21.0 – 24.0 °C
CO₂ 742 ppm
Flow 69.8 m³/s
OCCUPIED 111
(c) KPI tiles
OUTDOOR
23.6 °C E+ 23.8 66 %RH E+ 64
OUTDOOR PM2.5
169 µg/m³ E+ 160 Very unhealthy
GRID PRICE
0.960 $/kWh E+ 0.960 ON-PEAK
GRID CARBON
0.541 kg CO₂/kWh E+ 0.541
DEMAND RESPONSE
ACTIVE shed 300 kW
(d) Air-handling unit graphic (AHU-1)
RELIEF AIR
RETURN AIR ← zones
Return air CO₂ 613 ppm E+ 590 Δ+23.3
Return air temp 24.7 °C E+ 24.6 Δ+0.13
Return air RH 45 %RH E+ 43 Δ+1.8
RELIEF DMPR
OA temp (intake) 24.1 °C E+ 23.8 Δ+0.30
OA RH (intake) 67 %RH E+ 64 Δ+3.1
Mixed air temp 23.8 °C E+ 23.9 Δ−0.05
Filter ΔP 16 Pa E+ 10 Δ+6.4
Coil leaving air 9.3 °C E+ 9.7 Δ−0.36
Fan status ON
Supply air temp 12.8 °C E+ 10.1 Δ+2.73
SAT set point 12.8 °C
Supply air RH 98 %RH E+ 99 Δ−0.6
RETURN DMPR
Coil load (truth) 166 kW
OUTDOOR AIR
OA DAMPER
AFMS
MIXING
FILTER
PREHEAT
COOLING
SUPPLY FAN (VFD)
FRZ
SMOKE
SUPPLY AIR → VAV
OA damper cmd 100 %
OA damper pos 100 % E+ 100 Δ−0.1
OA flow 5.28 m³/s E+ 5.29 Δ−0.014
Preheat valve 0 % E+ 0 Δ+0.2
CHW valve 59 % E+ 59 Δ−0.1
HWS HWR
CHWS CHWR
Fan speed 31 % E+ 31 Δ+0.0
Fan power 2.7 kW E+ 2.6 Δ+0.07
Supply air flow 5.35 m³/s E+ 5.29 Δ+0.053
Duct static 374 Pa E+ 375 Δ−1.0
Static set point 375 Pa
Supply dew point 9.8 °C E+ 9.9 Δ−0.05